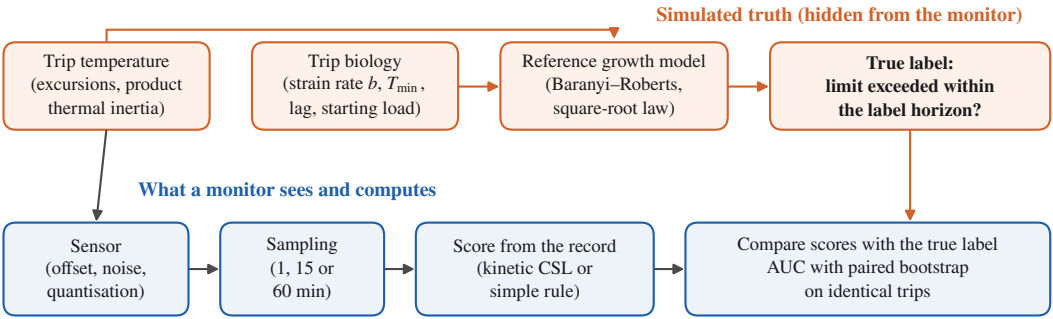
Simulated truth (hidden from the monitor)
Trip temperature (excursions, product thermal inertia)
Trip biology (strain rate b, $T_{\min}$, lag, starting load)
Reference growth model (Baranyi–Roberts, square-root law)
True label: limit exceeded within the label horizon?
What a monitor sees and computes
Sensor (offset, noise, quantisation)
Sampling (1, 15 or 60 min)
Score from the record (kinetic CSL or simple rule)
Compare scores with the true label AUC with paired bootstrap on identical trips